

Vincent Sun

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EDUCATION

University of Waterloo

Sep. 2023 – Apr. 2028

Bachelor of Computer Science

- cGPA: 91.18% (3.94/4.00), Awards: René Descartes National Scholarship, President's Scholarship of Distinction

TECHNICAL SKILLS

Certifications: AWS Certified Cloud Practitioner

Languages: Python, JavaScript, Typescript, Go, Java, BASH, C/C++, SQL

Frameworks/Libraries: React.js, Nest.js, Spring, Prisma, Django, React-Query, Knex, Node.js, Next.js, Express.js

Technologies: PostgreSQL, AWS, Docker, Kubernetes, Helm, MongoDB, Prometheus, Argo, Elasticsearch, REST

WORK EXPERIENCE

Cloud Developer Intern

Sep. 2025 – Present

SAP Inc. | Python, Go, Kubernetes, Docker, Helm, Prometheus, Argo

- Refactored **Helm** template in **Kubernetes** cluster to use multi-processing with parallel deployment tasks for rollout of new environment variables, improving release speed by **55%** and minimizing service interruptions
- Created **Prometheus** metrics to monitor and alert on **Argo** workflows, reducing incident detection time by **89%**

Software Engineering Intern

Jan. 2025 – Apr. 2025

MedMe Health Inc. (YC W21) | Typescript, Java, React.js, Nest.js, Spring, AWS, PostgreSQL

- Implemented event-driven **Twilio** email system built with **AWS SQS** and **Lambda** to replace **Graphile** job processing, reducing latency by over **60%** during high-traffic periods and improving error-handling using a DLQ
- Optimized **Spring** endpoint query speeds by **79%** through data loaders and all-or-none data transactions
- Added **Prometheus** metrics and built **5** custom dashboards to monitor **DynamoDB** usage with alert thresholds
- Created cookie consent management script and refactored **webpack** build process to block **100%** of third-party script injections, ensuring Quebec Law 25 compliance

Software Engineering Intern

May 2024 – Aug. 2024

Secoda Inc. (YC S21) | Typescript, Python, React, Django, Elasticsearch, PostgreSQL

- Engineered a ranking algorithm using dynamic weights with **Elasticsearch** to personalize user search results, leading to a reduction in average search rank of desired results by **23%** and increased click-through rate by **8%**
- Decreased integration query time by **47%** through optimization of **Django ORM** queries to a **Postgres** database
- Created analytics widgets for users to visualize their data trends and optimized metric update times by **32%**
- Developed a **Coalesce** integration to import jobs and tests built on **Snowflake** as well as data lineage

Software Developer

Jul. 2023 – Aug. 2023

Dupak Inc. | Google API

- Reduced annual audit preparation time by **80%** by developing a label-based organization method and implementing **Google API** plugins to automate metadata updates
- Continually updated and collaborated with management during development to ensure requirements were met

PROJECTS

LeetDance | *Typescript, Python, React.js, Next.js, Gemini, MediaPipe, OpenCV, FastAPI*

- Built a dance analysis platform for users to learn dancing by getting detailed comparisons to a reference video
- Dynamically analyzed videos using **Mediapipe** joint analysis and annotated videos with **OpenCV**
- Generated feedback to give to the user with **Gemini** and handled data transfer with **FastAPI**

Spectra | *Python, Raspberry Pi, OpenCV, Tensorflow*

- Created an over-the-shoulder bag with a camera that provides audio feedback to assist visually impaired people
- Integrated text recognition and emotion detection features using **OpenCV** and **Tensorflow** into a **Raspberry Pi**
- Demonstrated the product to Ontario Tech University and was recognized for incubation

Harmony Haven | *JavaScript, React.js, Node.js, Express.js, MongoDB, Chart.js*

- Developed a game-like web application to improve users' singing using real-time pitch comparisons
- Dynamically sourced songs and lyrics from **YouTube** and **Spotify API** to display to the user
- Analyzed pitch with the **Web Audio API**, visualized data with **Chart.js**, and stored information in **MongoDB**